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VML24CS071
S4 CSE-D ROLL NO:14

ASSIGNMENT-1

1.Write a C program priority.c thatimplementsnon-preemptive priority CPUscheduling (high value has higher priority).

Input Format:
n
PID1 AT1 BT1 PR1
PID2 AT2 BT2 PR2
.....
PIDn ATn BTn PRn
Sample Input:
5

P1 0 5 2
P2 0 4 5
P3 1 3 1
P4 2 3 4
P5 3 3 3
Sample Output:
Waiting Time:
P1 10
P2 0
P3 14
P4 2
P5 4
Turnaround Time:
P1 15
P2 4
P3 17
P4 5
P5 7
Average Waiting Time: 6.00
Average Turnaround Time: 9.60

PROGRAM

main.c

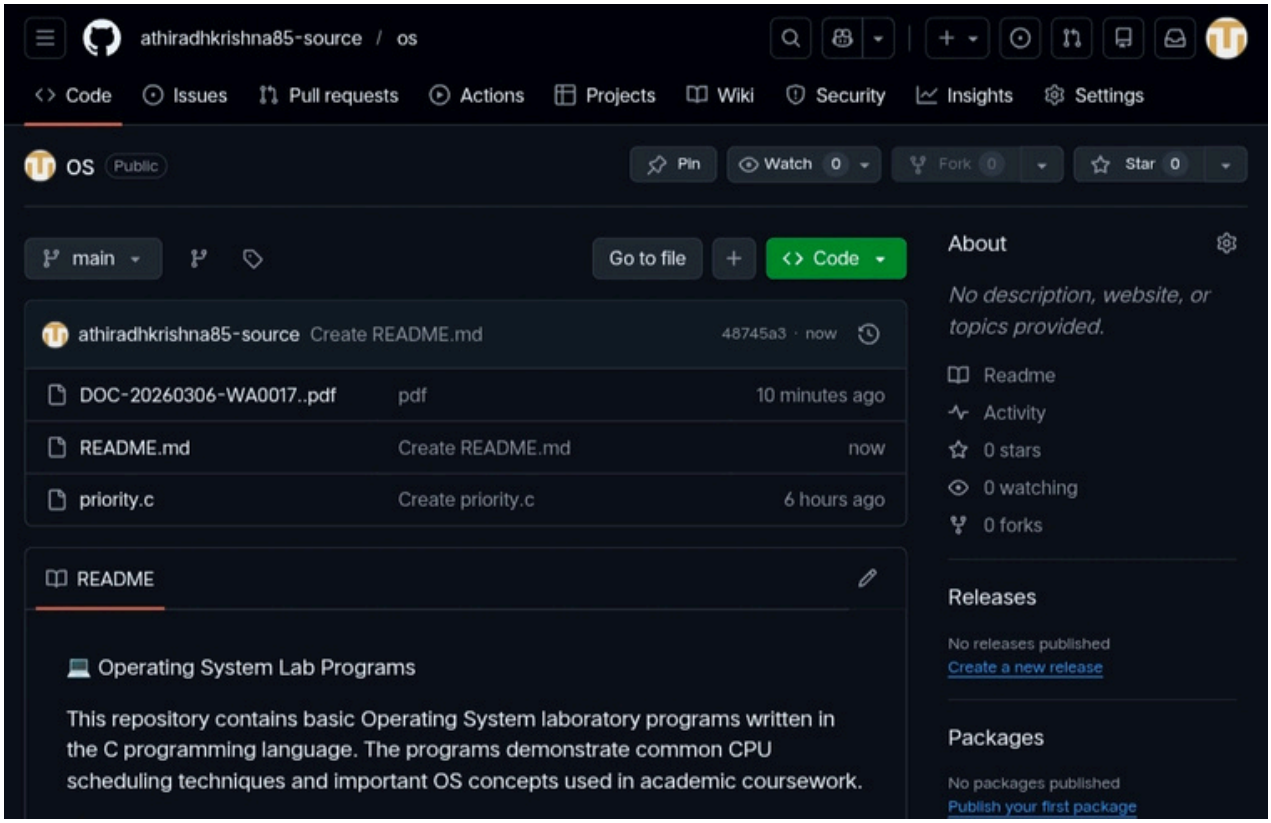
```c
#include <stdio.h>
#include <string.h>

struct process {
    char pid[10];
    int at, bt, pr;
    int wt, tat, ct;
    int completed;
};

int main() {
    int n, time = 0, completed = 0;
    float total_wt = 0, total_tat = 0;
    struct process p[20];

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        printf("Enter Process ID, Arrival Time,Time and Priority for process %d: ", i + 1);
        scanf("%s %d %d %d", p[i].pid, &p[i].at, &p[i].bt, &p[i].pr);
        p[i].completed = 0;
    }

    while (completed < n) {
        int idx = -1;
        int highest = -1;

        for (int i = 0; i < n; i++) {
            if (p[i].at <= time && p[i].completed == 0) {
                if (p[i].pr > highest) {
                    highest = p[i].pr;
                    idx = i;
                }
            }
        }

        if (idx != -1) {
            time += p[idx].bt;
            p[idx].ct = time;
            p[idx].tat = p[idx].ct - p[idx].at;
            p[idx].wt = p[idx].tat - p[idx].bt;

            total_wt += p[idx].wt;
            total_tat += p[idx].tat;

            p[idx].completed = 1;
            completed++;
        } else {
            time++;
        }
    }

    printf("Waiting Time:\n");
    for (int i = 0; i < n; i++) {
        printf("%s %d\n", p[i].pid, p[i].wt);
    }

    printf("Turnaround Time:\n");
    for (int i = 0; i < n; i++) {
        printf("%s %d\n", p[i].pid, p[i].tat);
    }

    printf("Average Waiting Time: %.2f\n", total_wt / n);
    printf("Average Turnaround Time: %.2f\n", total_tat / n);

    return 0;
}
```

OUTPUT

```
~ $ clang priority.c
~ $ ./a.out
Enter the number of processes: 5
Enter Process ID, Arrival Time, Burst Time and Priority
for process 1: p1 0 5 2
Enter Process ID, Arrival Time, Burst Time and Priority
for process 2: p2 0 4 5
Enter Process ID, Arrival Time, Burst Time and Priority
for process 3: p3 1 3 1
Enter Process ID, Arrival Time, Burst Time and Priority
for process 4: p4 2 3 4
Enter Process ID, Arrival Time, Burst Time and Priority
for process 5: p5 3 3 3
Waiting Time:
p1 10
p2 0
p3 14
p4 2
p5 4
Turnaround Time:
p1 15
p2 4
p3 17
p4 5
p5 7
Average Waiting Time: 6.00
Average Turnaround Time: 9.60
~ $
```

```
1:33 m ▾ ▣ ▣ •                    ⁸⁸ 2.33 KB/s ᵢₗₗ 18%
Report issues at https://bugs.termux.com
~ $ ^C
~ $ ^C
~ $ clang priority.c
~ $ ./a.out
Enter the number of processes: 5
Enter Process ID, Arrival Time, Burst Time and Priority
for process 1: p1 0 5 2
Enter Process ID, Arrival Time, Burst Time and Priority
for process 2: p2 0 4 5
Enter Process ID, Arrival Time, Burst Time and Priority
for process 3: p3 1 3 1
Enter Process ID, Arrival Time, Burst Time and Priority
for process 4: p4 2 3 4
Enter Process ID, Arrival Time, Burst Time and Priority
for process 5: p5 3 3 3
Waiting Time:
p1 10
p2 0
p3 14
p4 2
p5 4
Turnaround Time:
p1 15
p2 4
p3 17
p4 5
p5 7
Average Waiting Time: 6.00
Average Turnaround Time: 9.60
~ $
```

GETHUB



https://github.com/athiradhkrishna85-source/os/tree/main